Stroke

ORIGINAL CONTRIBUTION

Intracranial Hemorrhage Patterns and Outcomes in Minor Stroke: Analysis of the TEMPO-2 Trial

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BACKGROUND: Intracranial hemorrhage (ICH) negatively impacts functional outcomes after ischemic stroke, with potentially disproportionate impacts in patients with minor stroke. This study aimed to evaluate the effect of ICH on outcomes in minor ischemic stroke and to identify predictors associated with ICH.

METHODS: This was a secondary analysis of the TEMPO-2 multicenter, randomized trial, which compared tenecteplase with nonthrombolytic standard care in patients within 12 hours of symptom onset with minor stroke (National Institutes of Health Stroke Scale score ≤5) and a visible vessel occlusion or perfusion mismatch. Follow-up imaging was assessed for hemorrhage using the Heidelberg classification. Symptomatic ICH was defined as any hemorrhage associated with neurological deterioration. The primary outcome was return to premorbid functional status at 90 days, measured using the modified Rankin Scale. Mixed-effects regression was used to assess the effect of ICH status on outcomes, adjusting for treatment, age, sex, baseline stroke severity, and onset-to-randomization time, with region included as a random effect.

RESULTS: Among 884 participants, 865 had complete 24-hour imaging and follow-up. Using complete case analysis (n=865), any ICH occurred in 102 participants (11.8%). Patients with any ICH (median age, 70 years; 35.3% females) more frequently had premorbid hypertension (71.6% versus 57.7%) and atrial fibrillation (28.4% versus 18.1%). Any ICH was not associated with reduced odds of returning to baseline neurological function (adjusted odds ratio, 0.93 [95% CI, 0.87–1.00]) but was associated with higher 90-day mortality (9.8% versus 1.8%; adjusted hazard ratio, 3.71 [95% CI, 1.54–8.95]). Rates of any ICH were higher in tenecteplase than standard care (14.4% versus 9.2%; *P*=0.02), although most were petechial hemorrhagic transformations. Symptomatic ICH rates, although numerically higher, were not significantly different between the tenecteplase versus control arms (8 [1.9%] versus 2 [0.5%]; *P*=0.06).

CONCLUSIONS: Although most hemorrhages were minor, the presence of any ICH was strongly associated with increased mortality, highlighting that even minor hemorrhagic transformation may be prognostically significant in patients with minor ischemic stroke.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: atrial fibrillation ■ hemorrhage ■ infarction ■ perfusion ■ tenecteplase

More than half of all patients with ischemic stroke present with relatively mild symptoms, often defined as a National Institutes of Health Stroke Scale (NIHSS) score of ≤5.[1–3] Despite mild symptoms at presentation, unfavorable outcomes are common at 90 days, including disability in 29%, stroke recurrence in 10% and death in 4%.[4–9]

A feared complication of ischemia with or without reperfusion therapy is the development of intracranial hemorrhage (ICH).[10] Recent evidence indicates that

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Nonstandard Abbreviations and Acronyms

aRR	adjusted risk ratio
CONSORT	Consolidated Standards of Reporting Trials
ECASS II	European Cooperative Acute Stroke Study 2
ICH	intracranial hemorrhage
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale
NINDS	National Institute of Neurological Disorders and Stroke
SITS-MOST	Safe Implementation of Thrombolysis in Stroke Monitoring Study

even asymptomatic ICH (ie, hemorrhagic not accompanied by a neurological worsening) is independently associated with worse 90-day functional outcome and higher mortality.[11,12] As such, the risk of ICH must be carefully considered when offering hyperacute stroke treatment, and multiple predictive scores have been developed to estimate the risk of hemorrhage after thrombolysis.[13–24] In patients with minor stroke, the development of ICH may carry especially large functional implications. Given a relatively preserved baseline neurological status, even a small hemorrhagic event might tip the balance toward disability or dependency. Nonetheless, the major hemorrhage-risk prediction tools are largely developed and validated in populations with more severe strokes and have not been widely examined in populations with minor strokes.

In this post hoc analysis of TEMPO-2 (a randomized controlled trial of tenecteplase versus standard of care for minor ischemic stroke with proven occlusion), we aim to (1) describe the incidence and patterns of ICH including location, number, and radiological severity according to the Heidelberg bleeding classification and (2) to evaluate the association of these hemorrhagic characteristics with 90-day functional outcome. We explore associations between baseline clinical and imaging variables, treatment factors and risk of ICH development.

METHODS

Data can be made available to others on reasonable request and after signing appropriate data sharing agreements. Please send data access requests to scoutts@ucalgary.ca. Such requests must be approved by all the respective ethics boards and appropriate data custodians.

Study Design

This is a post hoc secondary analysis of the TEMPO-2 trial, a prospective, multicenter, open-label randomized phase 3 clinical trial with blinded outcome assessment evaluating intravenous tenecteplase (0.25 mg/kg) versus standard care in patients with minor stroke (NIHSS score ≤5) with visible occlusion and perfusion mismatch presenting within 12 hours of symptom onset. The protocol[25] and main results[26] have been published previously. In summary, the included patients were adults ≥18 years, independent at baseline (modified Rankin Scale [mRS] score ≤2), presenting with minor deficits (NIHSS score ≤5). Exclusion criteria included any contraindications to thrombolysis or if the treating physician deemed thrombolysis was indicated as part of the clinical standard of care. The results of this analysis are reported according to the CONSORT 2025 guidelines (Consolidated Standards of Reporting Trials).

Population

The intention-to-treat population from TEMPO-2 was analyzed using a complete case approach to identify any patients with ICH on routine follow-up imaging at 24 hours. Data were not imputed. These patients with any ICH on routine follow-up imaging were compared with those without any ICH. Baseline characteristics were compared between patients with any ICH versus patients with no ICH.

Outcomes

The primary outcome was return to baseline functional status using blinded mRS at 90 days using a sliding dichotomy approach. A responder was defined as follows: (1) if the premorbid mRS score is 0 to 1, then the mRS score, 0 to 1 at 90 days, was a responder (good outcome), or (2) if the premorbid mRS score is 2, then the mRS score of 0 to 2 was a responder (good outcome). Secondary outcomes included NIHSS score ≤5 at 5 days or discharge (whichever is sooner), Lawton Instrumental Activities of Daily Living at 90 days, quality of life measured at 90 days on the European quality of life score, 5 dimensions, 5 levels, recanalization status, proportion of patients receiving rescue endovascular thrombectomy and mortality at 90 days. All outcomes were assessed by an investigator blinded to treatment allocation.

Imaging Analyses

Hemorrhage grade on imaging was assessed using the Heidelberg bleeding classifications at 24-hour, routine, follow-up computed tomography imaging by a central core lab blinded to allocation and outcomes. The Heidelberg classifications differentiate hemorrhagic infarction without space-occupying effect (hemorrhagic infarction type 1 and hemorrhagic infarction type 2) and parenchymal hematoma with space-occupying effect (parenchymal hematoma type 1 and parenchymal hematoma type 2).[27,28] Other intracranial bleeding was reported, including intraventricular hemorrhage, subarachnoid hemorrhage, and subdural hemorrhage. The location of any hemorrhage was divided into 4 groups: subarachnoid hemorrhage, subdural, intraventricular, and parenchymal. Additionally, the side of the hemorrhage (left, right, or bilateral) and the number of hemorrhages (single or multiple) were recorded. Symptomatic ICH was defined as any hemorrhage temporally related to neurological worsening. Clinical worsening was defined by the NIHSS score worsening by a minimum of ≥2 points from baseline.

Privacy and Ethics

The TEMPO-2 trial was conducted in 48 sites across 10 countries internationally. Local ethics board approval was obtained at each participating site. The trial was regulated by Health Canada, with each site following its respective country's guidelines as required.

Statistical Analyses

Baseline characteristics and imaging variables were compared between patients with any ICH versus no ICH using descriptive statistics. ICH refers to ICH of any kind, which includes parenchymal hemorrhage, intraventricular hemorrhage, subarachnoid hemorrhage and subdural hemorrhage. These groups were further stratified by treatment type (intravenous tenecteplase versus standard of care) and compared using descriptive statistics. Categorical variables are expressed as frequencies and percentages, and quantitative variables as medians and interquartile ranges. Mixed-effects Poisson regression and mixed-effects linear regression were used to assess the effect of ICH status on binary and continuous outcomes, adjusting for age, sex, baseline stroke severity, and onset-to-randomization time, with region included as a random effect to account for clustering. Multivariable Poisson regression with robust error variance was used to identify independent predictors of ICH. Candidate variables were selected a priori based on clinical relevance and included demographics, vascular risk factors, baseline stroke characteristics, laboratory values, and treatment assignment; adjusted risk ratios (aRRs) with 95% CIs were reported.

Statistical significance was defined as 2-tailed $P<0.05$. All analyses were performed using Stata/MP version 17.0 (Stata Corp).

RESULTS

Baseline Characteristics in Patients With Versus Without ICH

Among 884 participants enrolled and randomized, 865 (97.8%) had available imaging at 24-hours and completed mRS at 90 days (Figure 1). There were 102 (11.8%) with any ICH on repeat imaging at 24-hours (40 [39.2%] in the control arm and 62 [60.8%] in the tenecteplase arm) as compared with 765 (88.2%) with no ICH (394 [51.6%] control, 369 [48.4%] tenecteplase arm). Baseline characteristics comparing patients with any ICH versus no ICH were largely similar, except that those patients with any ICH had higher rates of premorbid hypertension (73 [71.6%] versus 440 [57.7%]) and atrial fibrillation (29 [28.4%] versus 138 [18.1%]; Table 1). There was no statistically significant difference between the percent of recanalization achieved between the no ICH group (34.5%) and the ICH group (39.1%, P=0.49; Table 1). Rates of rescue EVT were significantly higher in the ICH group (7.8% versus 2.2%, P=0.0055; Table 1).

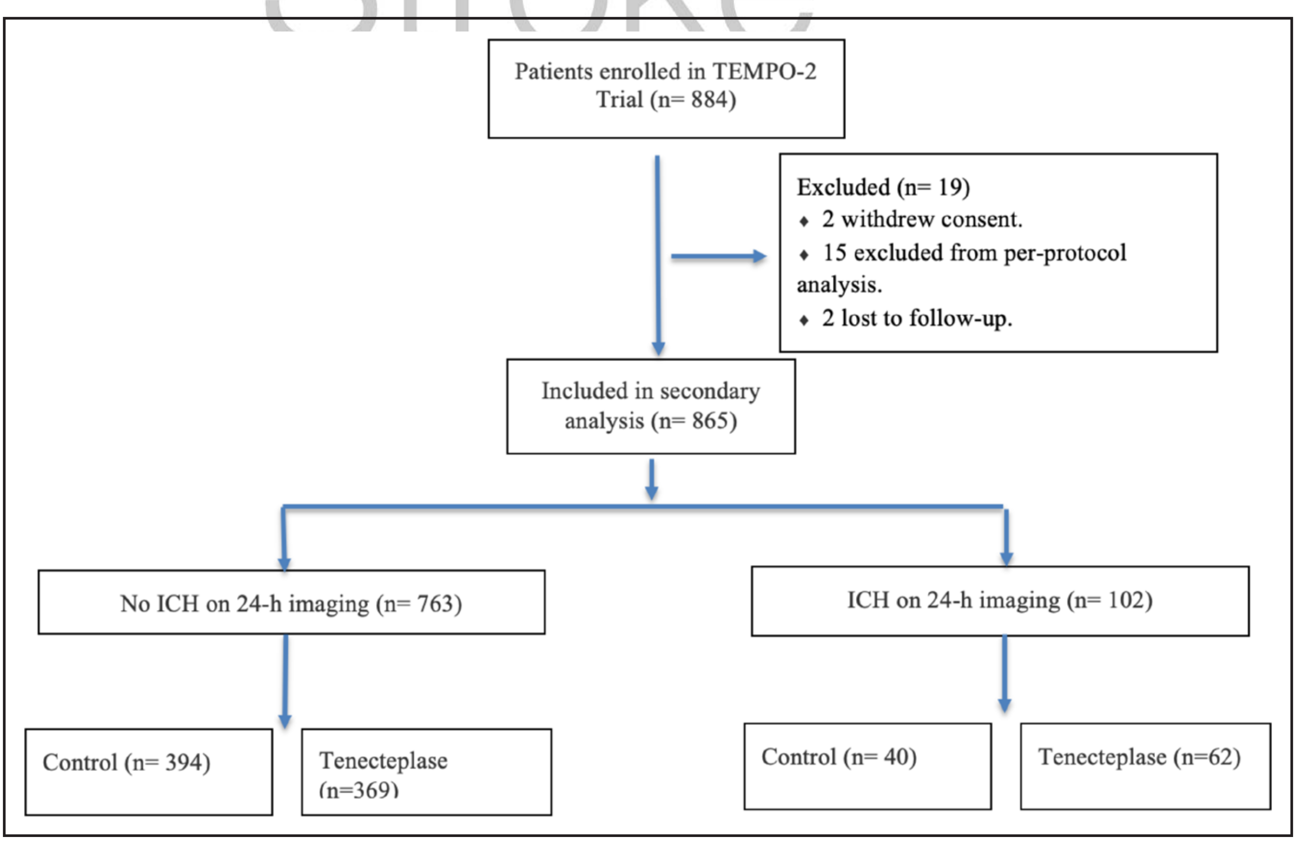

Figure 1. Flow chart of included patients in analysis.
ICH indicates intracranial hemorrhage.

Functional Outcomes in Patients With Versus Without ICH

There was no difference in responder rate between patients with ICH (71 [69.6%]) versus no ICH (566 [74.2%]; aRR, 0.93 [95% CI, 0.87–1.00]; Table 2). However, patients with ICH had significantly higher mortality at 90 days (10 [9.8%]) versus those who did not (14 [1.83%]; adjusted hazard ratio, 3.71 [95% CI, 1.54–8.95]). From serious adverse events, it was found that in the first 6 deaths were attributed to symptomatic ICH, 5 in the tenecteplase group, and 1 in the control. Of those in tenecteplase, 4 were parenchymal hemorrhage, and 1 was remote ICH. In the control group, there was 1 death attributed to subdural hemorrhage (Table S1). Deaths that occurred later did not appear to have a clear association with ICH. Additionally, patients with ICH were less likely to achieve an mRS score of 0 to 1 at 90 days compared with controls (aRR, 0.92 [95% CI, 0.88–0.97]; Table 2) and return to baseline function was lower in patients with ICH (aRR, 0.77 [95% CI, 0.69–0.85]; Table 2). All other secondary outcomes were similar in both groups (Table 2). Figure 2 depicts the mRS of patients divided by Heidelberg classification and further by treatment arm.

Comparison of ICH Patterns Between Tenecteplase and Control Arm

In patients with ICH, baseline characteristics were similar between tenecteplase versus control except more patients in the tenecteplase arm were women (26 [41.9%] versus 10 [25.0%]), had renal disease (7 [11.3%] versus 0 [0.0%]), higher NIHSS score at presentation (3 [1–4] versus 1 [0–3]), and prolonged onset to hospital arrival time (160 [73–391] versus 116 [68–219] minutes; Table S2).

Among patients with any ICH, more than half of them were hemorrhagic infarction type 1 and hemorrhagic infarction type 2 in both arms (control group 30 [6.91%] versus tenecteplase 37 [8.58%], Table 3; Figures 2 and 3). Within the ICH subgroup, rates of any ICH were significantly higher in the tenecteplase versus control arm (62 [14.4%] versus 40 [9.2%]; P=0.020). Symptomatic ICH rates were numerically higher but not statistically significant in the tenecteplase arm compared with the control arm (8 [1.9%] versus 2 [0.5%]; P=0.063; Table 3). There was no difference in single versus multiple ICH occurrence between each arm (Table 3).

Predictors of ICH

Multivariable analysis identified several independent factors associated with ICH in minor stroke patients (Table 4). Treatment with tenecteplase was associated with a higher risk of ICH compared with standard care (aRR, 1.51 [95% CI, 1.01–2.26]). Hypertension (aRR, 1.82 [95% CI, 1.13–2.92]) and atrial fibrillation (aRR,

Table 1. Baseline Characteristics in Patients Who Had Any Hemorrhage Versus Those Who Did Not on 24-Hour Follow-Up Imaging

	No ICH	ICH	*P* value
N	765 (88.2%)	102 (11.8%)	
Age, y	72 (61–80)	70 (62–77)	0.28
Sex, female	326 (42.6%)	36 (35.3%)	0.17
Race			
White	657 (85.9%)	80 (78.4%)	0.23
Asian	67 (8.8%)	15 (14.7%)	
Other	21 (2.7%)	4 (3.9%)	
Hispanic	14 (1.8%)	0 (0.0%)	
Black	12 (1.6%)	1 (1.0%)	
First Nations	7 (0.9%)	2 (2.0%)	
Pacific Islander	1 (0.1%)	0 (0.0%)	
Hypertension	440 (57.5%)	73 (71.6%)	0.01
Past smoking	296 (38.7%)	43 (42.2%)	0.52
Hyperlipidemia	307 (40.1%)	38 (37.3%)	0.59
Diabetes	147 (19.2%)	19 (18.6%)	1.00
Past stroke	131 (17.1%)	22 (21.6%)	0.27
Atrial fibrillation	138 (18.0%)	29 (28.4%)	0.02
Ischemic heart disease	120 (15.7%)	18 (17.6%)	0.66
Congestive heart failure	29 (3.8%)	5 (4.9%)	0.58
Chronic renal disease	31 (4.1%)	7 (6.9%)	0.20
Peripheral vascular disease	25 (3.3%)	2 (2.0%)	0.76
Past ICH	3 (0.4%)	1 (1.0%)	0.39
mRS score at baseline			
0	599 (78.3%)	79 (77.5%)	0.78
1	105 (13.7%)	13 (12.7%)	
2	61 (8.0%)	10 (9.8%)	
NIHSS score at baseline	2 (1–3)	2 (1–3)	0.53
Hemoglobin, g/L	140 (130–150)	141 (130–153)	0.65
Glucose, mmol/L	6 (6–7)	6 (6–8)	0.38
Creatinine, μmol/L	82 (69–97)	89 (74–103)	<0.0010
ASPECTS baseline	10 (10–10)	10 (9–10)	0.00
Onset to randomization time	277 (161–449)	294 (177–436)	0.81
Onset to ED time	153 (74–337)	134 (71–326)	0.62
Onset to treatment time	300 (174–475)	310 (190–462)	0.82
Occlusion location at baseline			
LVO	87 (11.4%)	15 (14.7%)	0.30
MeVO	408 (53.5%)	62 (60.8%)	
VB	41 (5.4%)	3 (2.9%)	
Focal perfusion lesion	219 (28.7%)	22 (21.6%)	
None	8 (1.0%)	0 (0.0%)	
Recanalization achieved	152 (34.5%)	25 (39.1%)	0.49
Rescue EVT Performed	17 (2.2%)	8 (7.8%)	0.0055

Data is represented as n (%) or median (IQR). Race and ethnicity were self-reported; ethnicity was binary, with Hispanic or non-Hispanic being the options. Time reported in minutes. ASPECTS indicates Alberta Stroke Program Early Computed Tomography Score; ED, emergency department; ICH, intracranial hemorrhage; IQR, interquartile range; LVO, large vessel occlusion; MeVO, medium vessel occlusion; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; and VB, vertebrobasilar artery.

Table 2. Functional Outcomes at 90 Days in Patients With No ICH Versus ICH on 24-Hour Follow-Up Imaging

	No ICH	ICH	Risk difference (95% CI)	Unadjusted RR (95% CI)	Adjusted RR (95% CI)
Responder at 90 d	566 (74.2)	71 (69.6)	−0.05 (−0.14 to 0.05)	0.94 (0.82 to 1.07)	0.93 (0.87 to 1.00)
Secondary end point					
mRS score, 0–1 at 90 d	542 (71.0)	68 (66.7)	−0.04 (−0.14 to 0.05)	0.94 (0.81 to 1.08)	0.92 (0.88 to 0.97)
mRS score, 0–2 at 90 d	651 (85.3)	80 (78.4)	−0.07 (−0.15 to 0.01)	0.92 (0.83 to 1.02)	0.92 (0.81 to 1.04)
mRS score at 90 d–median	1 (0–2)	1 (0–2)			
NIHSS score of 0 at 5 d or discharge	421 (55.2)	45 (46.4)	−0.09 (−0.19 to 0.02)	0.84 (0.67 to 1.05)	0.83 (0.68 to 1.02)
Return to baseline function, Y/N	389 (50.8)	40 (39.2)	−0.12 (−0.22 to −0.02)	0.77 (0.60 to 0.99)	0.77 (0.69 to 0.85)
Lawton IADL score–mean	20.4 (4.9)	19.58 (6.2)	−0.81 (−1.92 to 0.29)		−1.01 (−2.60 to 0.58)
EQ-5D index–mean	0.84 (0.2)	0.74 (0.3)	−0.10 (−0.15 to −0.05)		−0.10 (−0.15 to −0.05)
EQ-5D VAS–mean	75.7 (20.0)	66.69 (28.5)	−9.01 (−13.5 to −4.5)		−8.83 (−13.42 to −4.24)
Mortality at 90 d	14 (1.83)	10 (9.8)	N/A	HR, 4.47 (1.87 to 10.65)	3.71 (1.54 to 7.98)

Data is reported as a number (percent) unless otherwise specified. Adjusted for age, sex at birth, baseline NIHSS score, onset to randomization time and treatment group. EQ-5D-5 indicates European quality of life score, 5 dimensions, 5 levels; HR, hazard ratio; ICH, intracranial hemorrhage; Lawton IADL, Lawton–Brody Instrumental Activities of Daily Living Scale index; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; RR, risk ratio; VAS, visual analogue scale; and Y/N, yes or no.

1.66 [95% CI, 1.06–2.59]) were significant predictors, as was higher creatinine, although the effect size was small (aRR, 1.00 [95% CI, 1.00–1.01]). Other factors, including sex, age, baseline NIHSS, glucose, large vessel occlusion, and onset-to-randomization time, did not demonstrate an independent association with ICH.

DISCUSSION

In the TEMPO-2 trial, the presence of any ICH at 24 hours was observed in a minority of patients (11.8%) with minor stroke and confirmed vessel occlusion or perfusion lesion, with a higher frequency in the tenecteplase group (14.4%) compared with the control (9.2%). Patients who developed ICH were more likely to have a history of hypertension, atrial fibrillation and to have received rescue EVT. Although the occurrence of ICH was associated with higher mortality at 90 days, recovery to baseline mRS and other secondary outcomes were similar between those with and without ICH. Within the ICH subgroup, symptomatic ICH was uncommon and did not differ significantly between treatment arms. Parenchymal

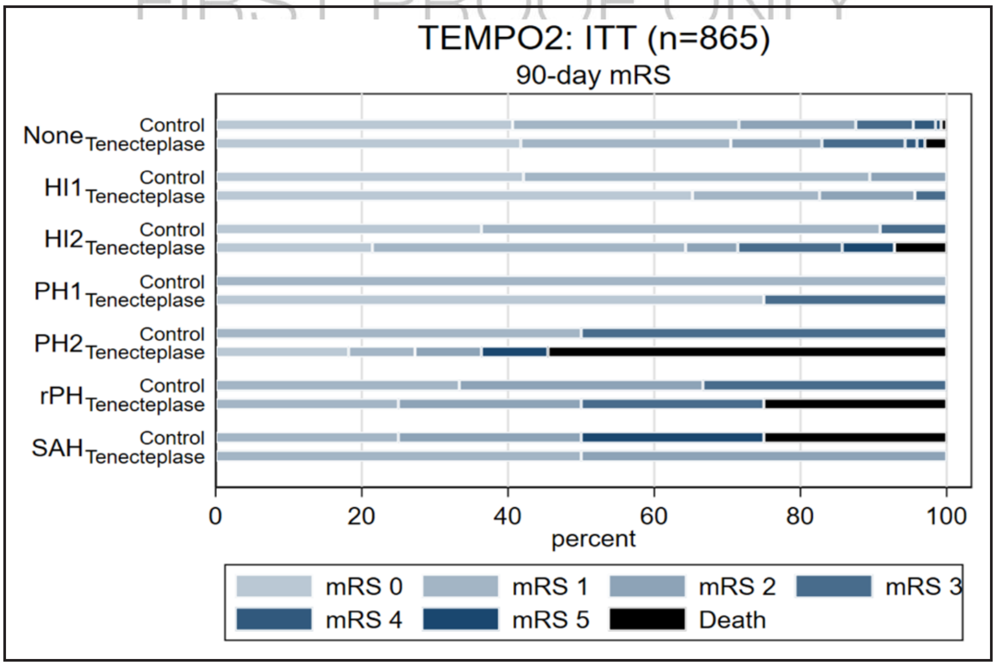

Figure 2. Hemorrhage grades in TEMPO-2 as per Heidelberg classification for the 865 patients with whom complete follow-up data at 90 days and follow-up imaging at 24 hours were collected.
HI1 indicates hemorrhagic infarction type 1; HI2, hemorrhagic infarction type 2; mRS, modified Rankin Scale; PH1, parenchymal hematoma type 1; PH2, parenchymal hematoma type 2; RPH, remote parenchymal hemorrhage; and SAH, subarachnoid hemorrhage.

Table 3. Comparison of Intracranial Hemorrhage Patterns on 24-Hour Follow-Up Imaging

	Control	Tenecteplase	*P* value
N	434 (50.2%)	431 (49.8%)	
Any hemorrhage	40 (9.2%)	62 (14.4%)	0.020
Symptomatic ICH	2 (0.5%)	8 (1.9%)	0.063
Location of hemorrhage			
Subarachnoid	5 (1.2%)	13 (3.0%)	0.060
Subdural	0 (0.0%)	1 (0.2%)	0.50
Intraventricular	0 (0.0%)	4 (0.9%)	0.061
Parenchymal	36 (8.3%)	59 (13.7%)	0.012
Heidelberg class			
HI1 (1a)	19 (4.4%)	23 (5.3%)	0.045
HI2 (1b)	11 (2.5%)	14 (3.2%)	
PH1 (1c)	1 (0.2%)	4 (0.9%)	
PH2 (2)	2 (0.5%)	11 (2.6%)	
RPH (3a)	3 (0.7%)	8 (1.9%)	
SAH (3c)	4 (0.9%)	2 (0.5%)	
ICH side			
Left	12 (33.3%)	24 (40.7%)	
Right	22 (61.1%)	32 (54.2%)	0.78
Bilateral	2 (5.6%)	3 (5.1%)	
ICH number			
Single	16 (44.4%)	22 (37.3%)	
Multiple	20 (55.6%)	37 (62.7%)	0.52

The Heidelberg classification also denotes SAH; however, the classification represents the most severe location. There were 4 cases of intraventricular hemorrhage (Heidelberg class 3b) and 1 case of subdural hemorrhage (Heidelberg class 3d). Each of these cases occurred concurrent with a PH type 2 hemorrhage (Heidelberg class 2), and all 5 were considered to be clinically symptomatic ICH. In addition to the 6 instances of isolated SAH in the table above, there were 12 additional occurrences of SAH cooccurring with other hemorrhage types. HI1 indicates hemorrhagic infarction type 1; H12, hemorrhagic Infarction type 2; ICH, intracranial hemorrhage; PH1, parenchymal hemorrhage type 1; PH2, parenchymal hemorrhage type 2; RPH, remote parenchymal hemorrhage; and SAH, subarachnoid hemorrhage.

hemorrhages were more frequent in the tenecteplase group, but the overall distribution of hemorrhage types was otherwise comparable across groups.

An additional observation was that rescue EVT occurred more frequently among participants with any ICH. This finding should be interpreted cautiously, as any ICH as defined in the current study was predominantly radiographic hemorrhagic transformation rather than symptomatic hemorrhage. Patients requiring rescue EVT likely represent a higher-risk subgroup with persistent occlusion or early clinical worsening, factors that are themselves associated with larger infarct burden and greater propensity for hemorrhagic transformation (confounding by indication). Moreover, petechial hemorrhagic transformation (hemorrhagic infarction type 1–hemorrhagic infarction type 2) has been described as a potential marker of reperfusion and early recanalization rather than a direct mediator of poor outcome in selected cohorts, which aligns

Table 4. Multivariable Predictors of ICH in Patients With Minor Stroke With Effect Size Represented as Adjusted Risk Ratio for Dichotomous Variables

Variable	Adjusted risk ratio	95% CI
Treatment: tenecteplase	1.51*	1.01–2.26
Age, y	0.98*	0.97–1.00
Sex: male	1.19	0.791.80
Baseline NIHSS	1.04	0.91–1.20
Large vessel occlusion†	1.37	0.78–2.40
Onset to randomization time	1.00	1.00–1.00
Hypertension	1.82*	1.13–2.92
Atrial fibrillation	1.66*	1.06–2.59
Creatinine, μmol/L	1.00*	1.00–1.01
Glucose, mmol/L	0.98	0.91–1.06

ICH indicates intracranial hemorrhage; and NIHSS, National Institutes of Health Stroke Scale.

*$P<0.05$.

†Intracranial internal carotid artery, M1 segment of the middle cerebral artery. M2 segment of the middle cerebral artery or distal A2 segment of the anterior cerebral artery. Intracranial vertebral artery, basilar artery or branches, posterior cerebral artery.

with prior work in proximal MCA occlusion treated with thrombolysis.[29] At the same time, reperfusion therapies including EVT are consistently associated with higher rates of radiographic hemorrhagic transformation across contemporary EVT data sets,[30–32] even when symptomatic ICH remains uncommon.

Previous studies have demonstrated that ICH following acute ischemic stroke is associated with worse long-term outcomes, even when not immediately symptomatic.[33] While early work, including the NINDS trials (National Institute of Neurological Disorders and Stroke), defined symptomatic ICH as any hemorrhage temporally related to neurological worsening,[34] this definition has since been recognized as overly inclusive and poorly correlated with long-term disability and mortality. More contemporary classification systems, such as the ECASS II (European Cooperative Acute Stroke Study 2) and modified version of the SITS-MOST criteria (Safe Implementation of Thrombolysis in Stroke Monitoring Study), better distinguish clinically meaningful hemorrhagic events. The SITS-MOST definition shows the strongest correlation with mortality, while ECASS II has higher interrater agreement.[34] These have now been codified in the Heidelberg bleeding classification, which has become the gold standard definition.[27] Importantly, subsequent studies have shown that even asymptomatic hemorrhage, or those that are not necessarily associated with acute clinical deterioration, can still be prognostically relevant, with asymptomatic hemorrhage being linked to poorer functional outcomes and higher mortality compared with patients without hemorrhage.[33,35] Additionally, studies have examined the type of hemorrhage using various classification systems and found variable correlation

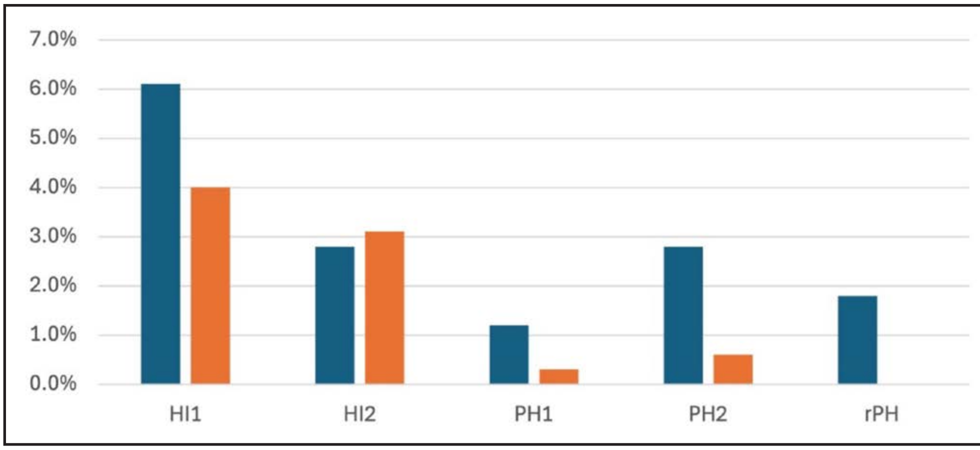

Figure 3. Bar graph depicting the incidence of intracranial hemorrhage (by Heidelberg classification) in tenecteplase (n=60) vs control (n=33) arms.

HI1 indicates hemorrhagic infarction type 1; HI2, hemorrhagic infarction type 2; PH1, parenchymal hematoma; PH2, parenchymal hematoma type 2; and RPH, remote parenchymal hematoma.

with functional outcome. There is a trend that hemorrhage involving noninfarcted tissue consistently correlates with poorer functional outcome.[36–38] However, some studies also find that more broadly, any type of hemorrhage of infarcted tissue is negatively correlated with functional outcome.[37,38] Collectively, there are multiple possible interpretations. One is that hemorrhage is indeed causative in poorer outcomes, something that appears obvious with symptomatic hemorrhage. There may be confounding factors, such as a delay in recommencing antithrombotics. Alternatively, hemorrhage is simply a marker of more severe neurological injury due to ischemia, and it is this underlying injury that is the reason for poorer outcomes.

This study has several strengths, including its randomized controlled design, multicenter participation, and the use of an independent core laboratory for imaging assessment, which enhances the validity and generalizability of the findings. However, several limitations should be acknowledged. The analysis was a post hoc subgroup evaluation that was not prespecified, and the relatively small number of patients with intracerebral hemorrhage limits statistical power to detect modest differences between treatment arms and results in wide CIs. Further analyses regarding specific hemorrhage categories were not pursued due to the relatively small sample. Infarct volume was not available, and this could also be a confounder associated with poor outcomes. Future studies should aim to examine the possible implications of medication initiation, hemorrhage volume and other potential related factors.

CONCLUSIONS

This secondary analysis of the TEMPO-2 trial demonstrates that the majority of hemorrhages in this population were radiographically minor. However, the occurrence of any ICH was associated with increased mortality and poorer functional recovery.

ARTICLE INFORMATION

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Affiliations

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Author Contributions

Conceptualization was performed by Drs Massicotte and Singh. Formal analysis was performed by Dr Massicotte, Dr Vatanpour, C. Doolan, J. Zhang, and Dr Singh. Supervision was conducted by Drs Singh, Coutts, and Hill. Writing the original draft was performed by Drs Massicotte (lead) and Singh (supporting). Writing review and editing were performed by all the authors.

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Supplemental Material

Tables S1–S2
CONSORT Checklist

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